

C++ Inheritance Programs Using Student Class

1. Single Inheritance

```
#include <iostream>
using namespace std;

class Student
{
public:
    string name;
    int rollNo;

    void getStudent()
    {
        cout << "Enter name: ";
        cin >> name;
        cout << "Enter roll no: ";
        cin >> rollNo;
    }
};

class Result : public Student
{
public:
    int marks;

    void getResult()
    {
        cout << "Enter marks: ";
        cin >> marks;
    }

    void display()
    {
        cout << "\nName: " << name;
        cout << "\nRoll No: " << rollNo;
        cout << "\nMarks: " << marks;
    }
};

int main()
{
    Result r;
    r.getStudent();
    r.getResult();
    r.display();
    return 0;
}
```

2. Multiple Inheritance

```
#include <iostream>
using namespace std;

class Student
{
public:
    string name;
    int rollNo;

    void getStudent()
    {
        cout << "Enter name: ";
        cin >> name;
        cout << "Enter roll no: ";
        cin >> rollNo;
    }
};

class Marks
{
public:
    int marks;

    void getMarks()
    {
        cout << "Enter marks: ";
        cin >> marks;
    }
};

class Result : public Student, public Marks
{
public:
    void display()
    {
        cout << "\nName: " << name;
        cout << "\nRoll No: " << rollNo;
        cout << "\nMarks: " << marks;
    }
};

int main()
{
    Result r;
    r.getStudent();
    r.getMarks();
    r.display();
    return 0;
}
```

3. Multilevel Inheritance

```
#include <iostream>
using namespace std;

class Student
{
public:
    string name;
    int rollNo;

    void getStudent()
    {
        cout << "Enter name: ";
        cin >> name;
        cout << "Enter roll no: ";
        cin >> rollNo;
    }
};

class Marks : public Student
{
public:
    int marks;

    void getMarks()
    {
        cout << "Enter marks: ";
        cin >> marks;
    }
};

class Result : public Marks
{
public:
    void display()
    {
        cout << "\nName: " << name;
        cout << "\nRoll No: " << rollNo;
        cout << "\nMarks: " << marks;
    }
};

int main()
{
    Result r;
    r.getStudent();
    r.getMarks();
    r.display();
    return 0;
}
```

Note: These are beginner-friendly C++ programs using Student class and can be directly compiled in a C++ compiler.